9na Clase 17/10/2025

# 75.43 Introducción a los Sistemas Distribuidos
# 95.60 Redes y Aplicaciones Distribuidas
# TA048 Redes

**Tema: Capa de enlace (I)**

Capítulo 6: desde el inicio hasta *6.3 Multiple Access Links and Protocols* inclusive, del libro *Computer Networking : A Top-Down Approach. James Kurose and Keith Ross. Publisher: Pearson Edition: 8th, 2021.*

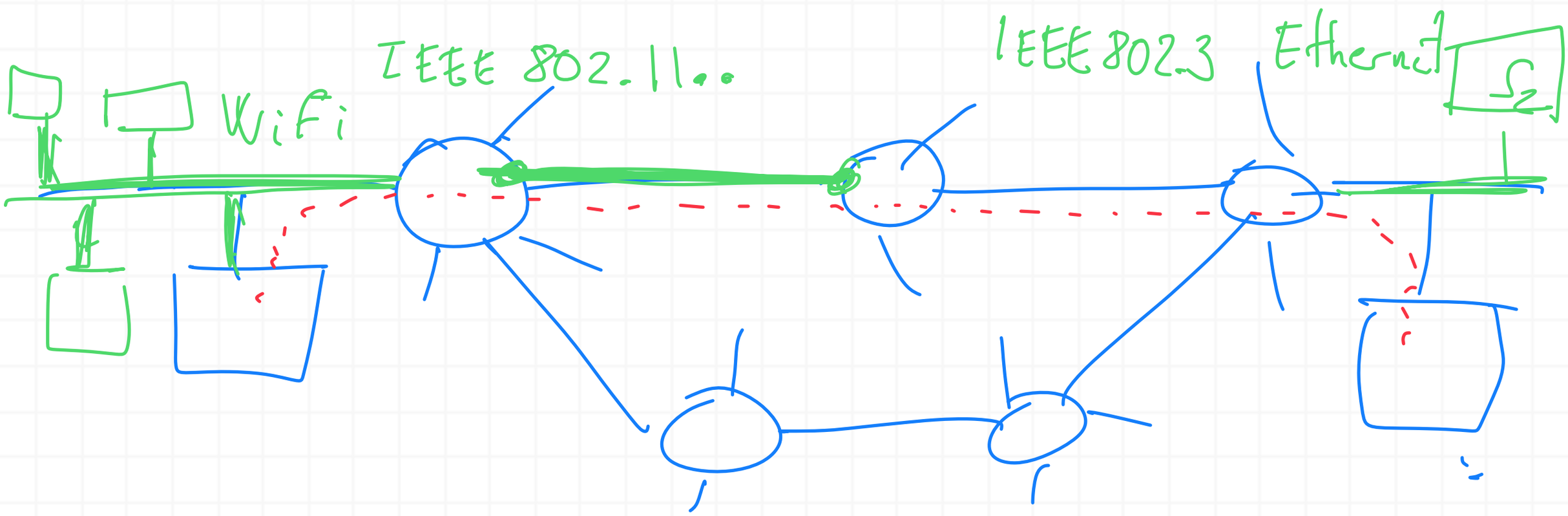
Dr. Ing. J. Ignacio Alvarez-Hamelin

# Clase de hoy

- Generalidades de la capa de enlace
- Medidas físicas en la capa de enlace
- Protocolos de acceso múltiple

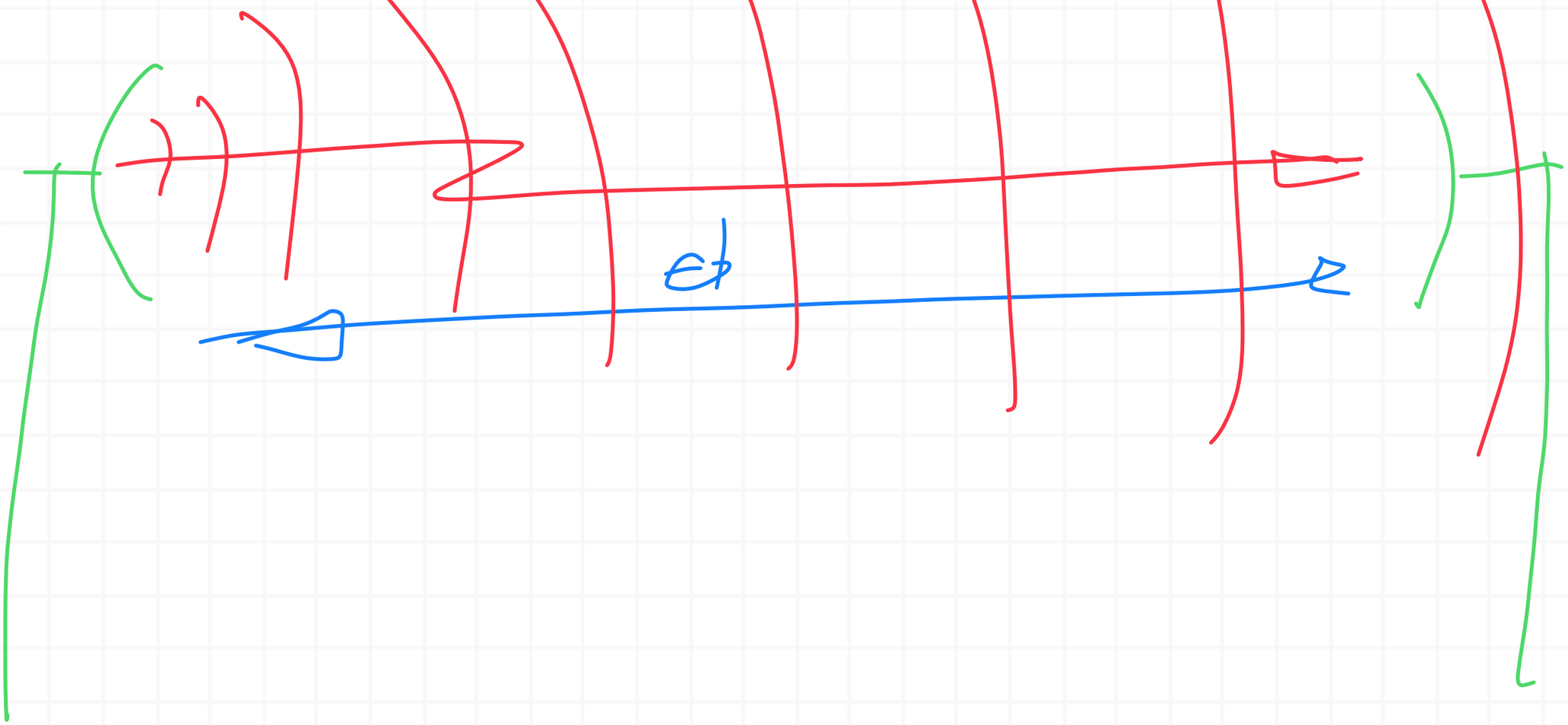
**Capa de enlace**

- Tipos: punto a punto y acceso múltiple
- Servicios:
  - Separación de datos (*framming*)
  - Acceso al enlace (MAC)
  - Entrega confiable
  - Detección y corrección de errores



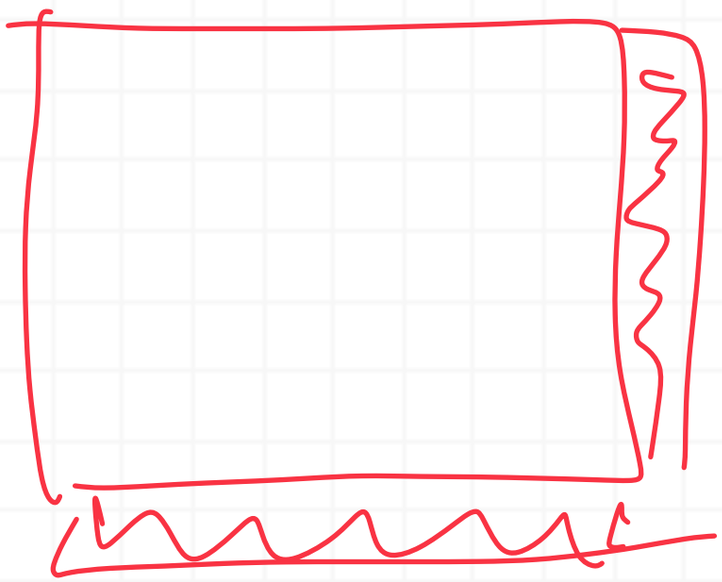
**Medidas físicas:**

- Potencia y atenuación
- BER (*Bit Error Rate*)
- Detección y Corrección de Errores
  - Verificación de paridad
    - Unidimensional
    - Bidimensional
  - CRC (*Cyclic Redundancy Check*)



$$BER = \frac{\text{Nº bits errados}}{\text{Nº total de bits transmitidos}}$$

CRC

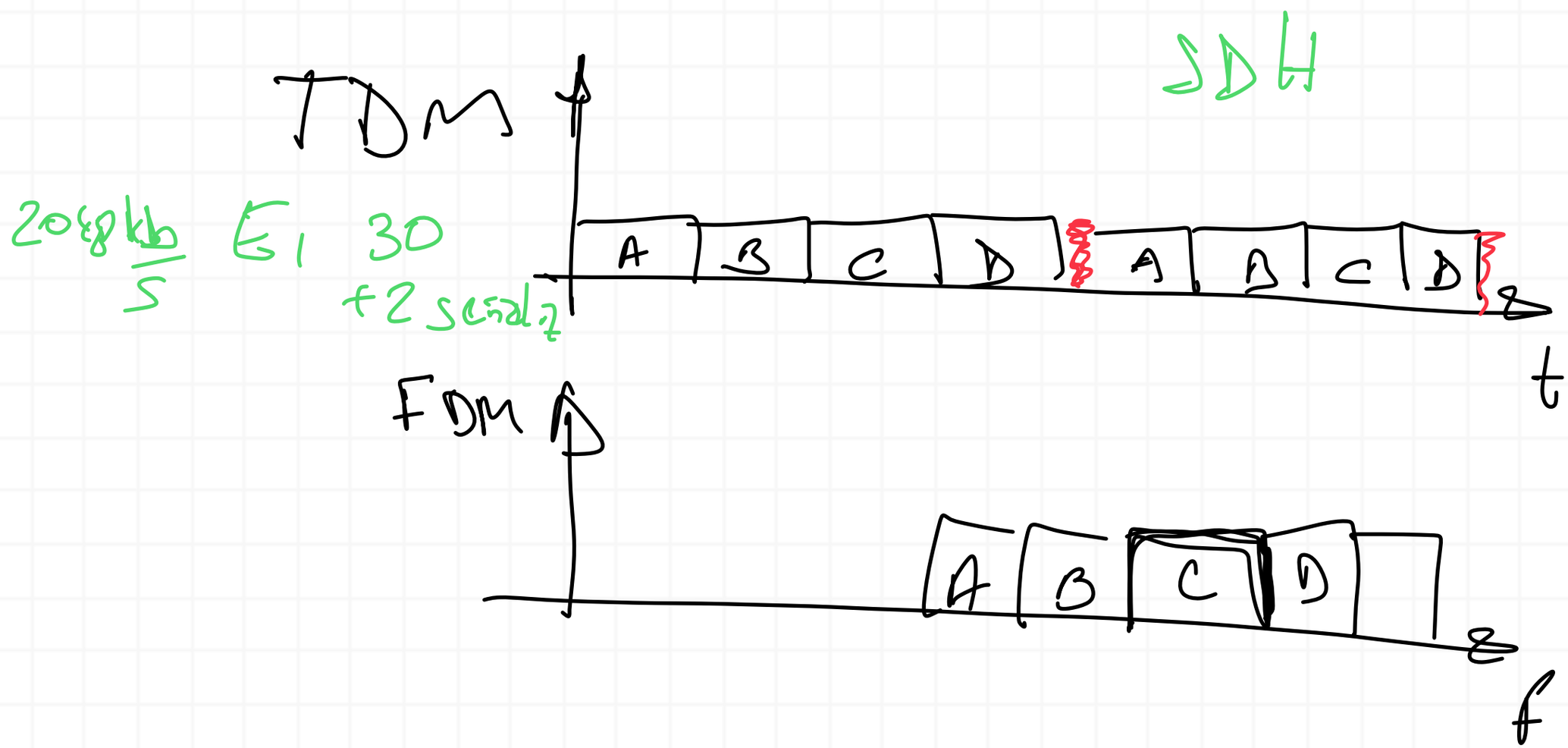


$$\widehat{BER} = Cod.\ Det.\ Err$$

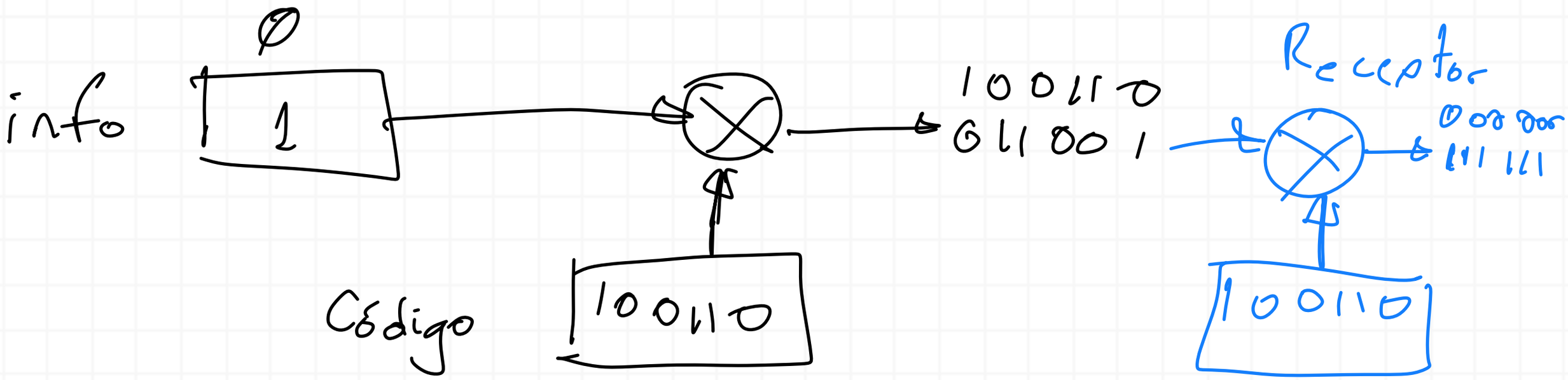
$$BER = 10^{-9}$$

## Protocolos de Acceso Múltiple

- Medios de difusión (*broadcasting* )
- Formas de administrar el uso del canal
  - Partición del canal (TDM, FDM, CDMA)
  - Acceso aleatorio (ALOHA, CSMA/CD, CSMA/CA)
  - Por turnos (*token* protocols*)

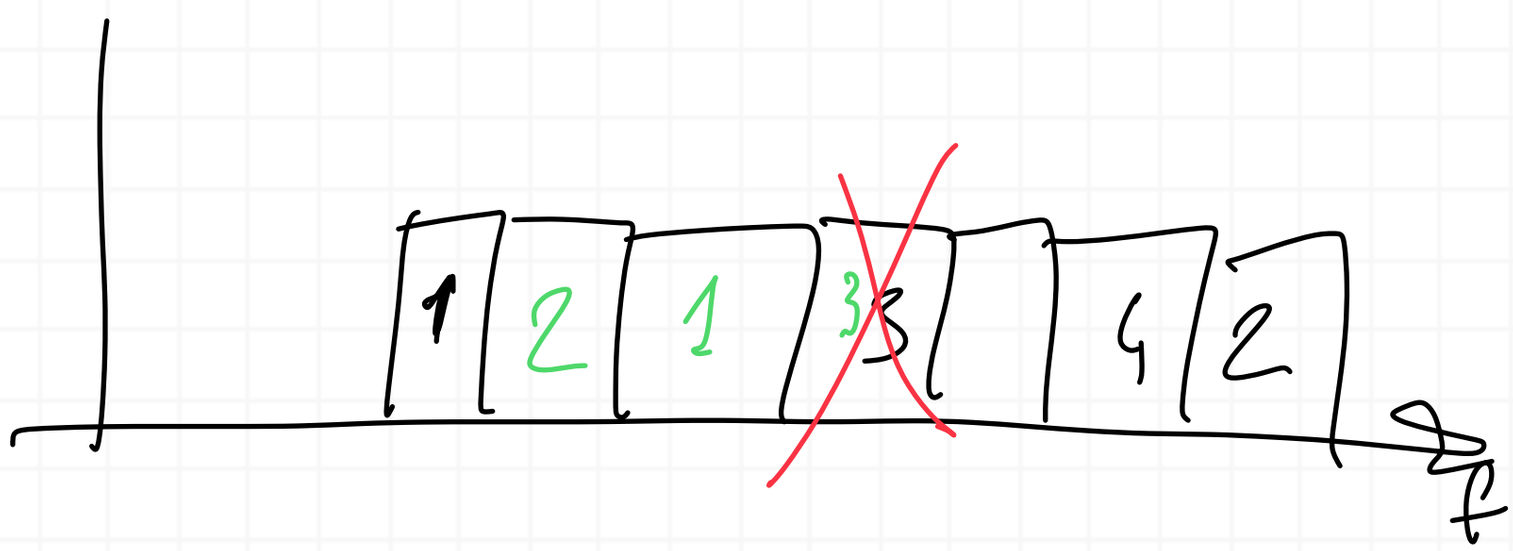


# CDMA

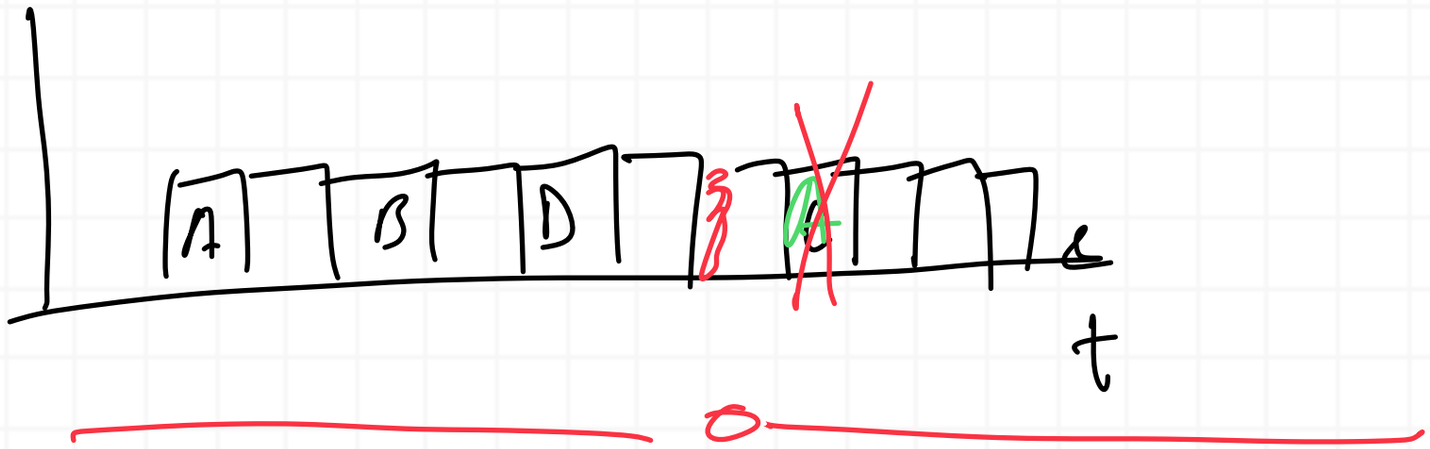



## Direct Sequence

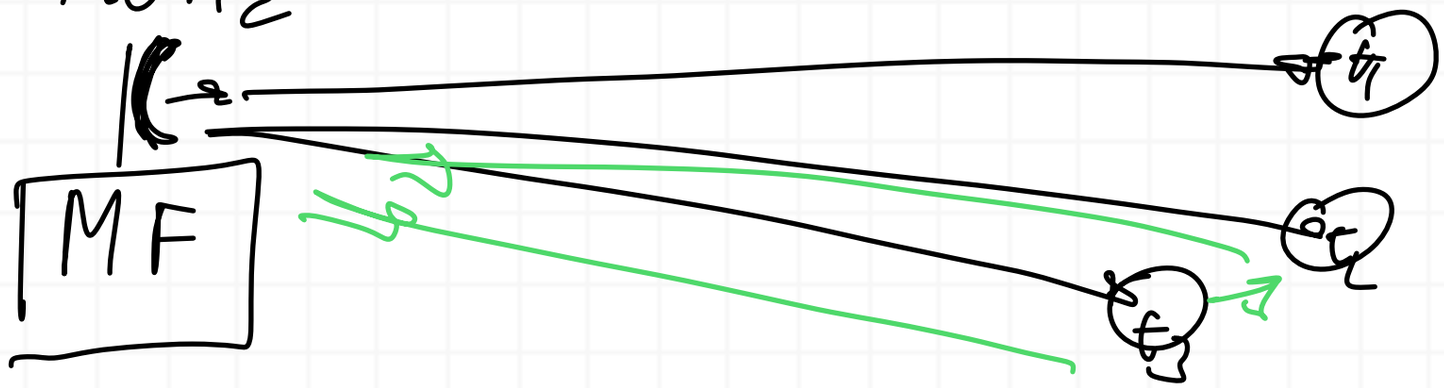
## Frequency Hopping

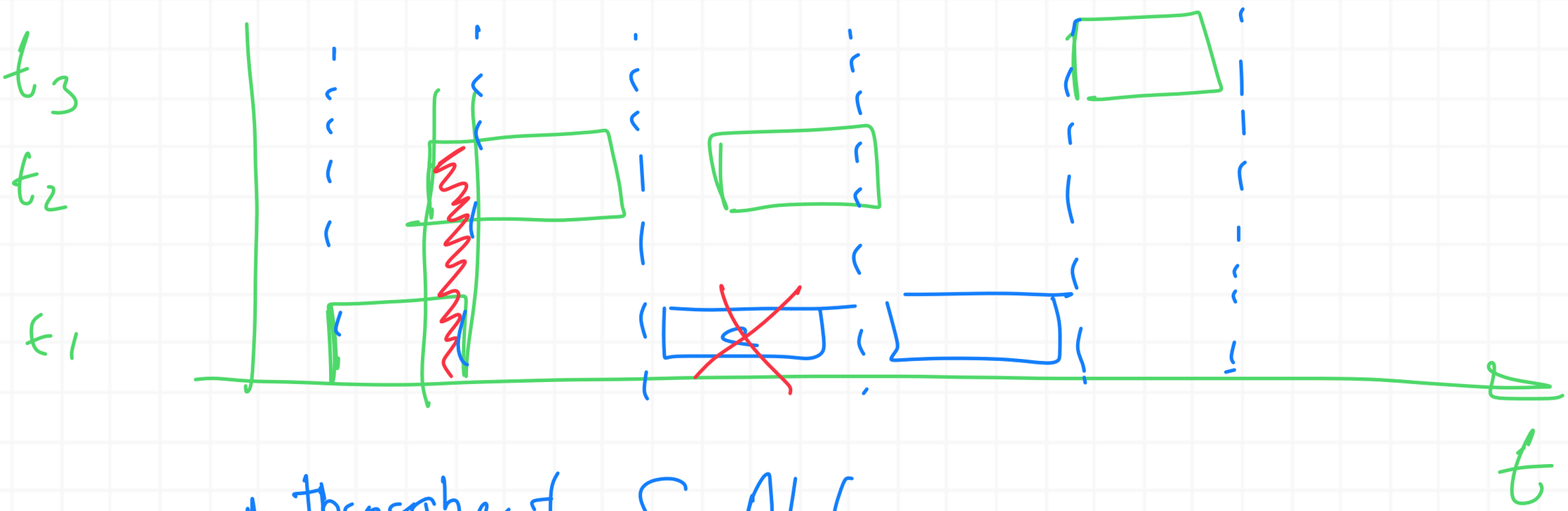


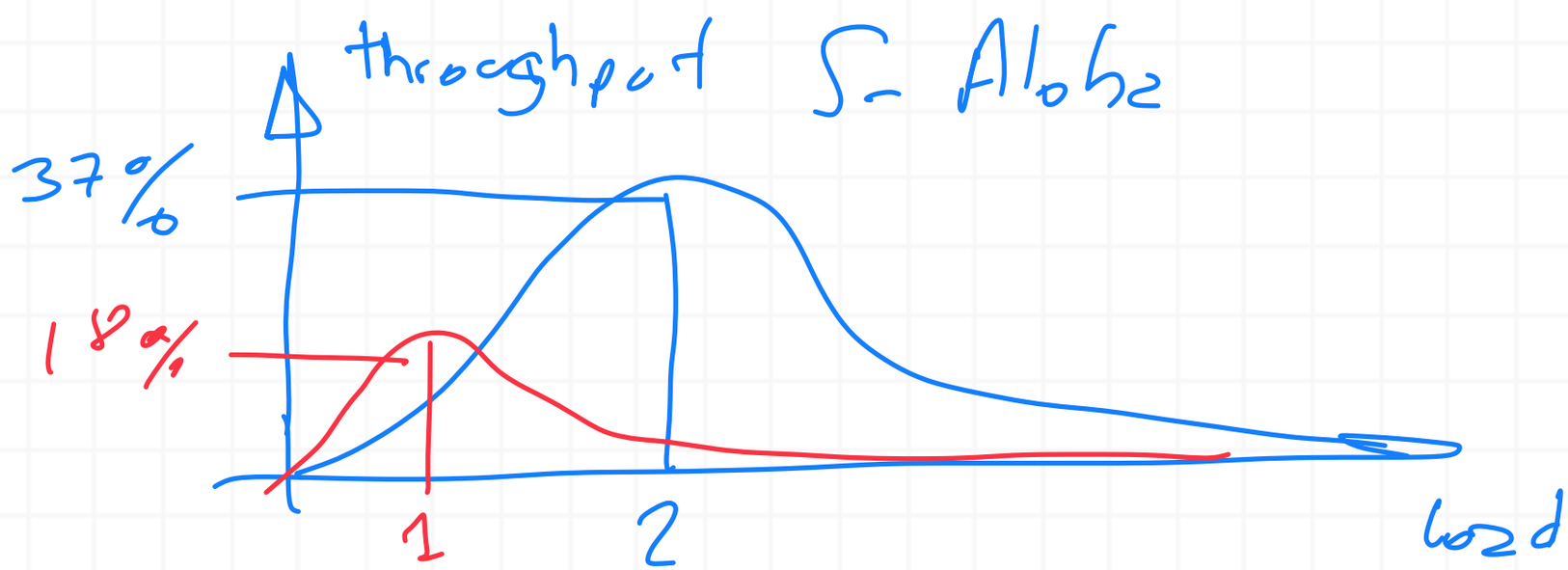
# Time Hopping



# Aloha



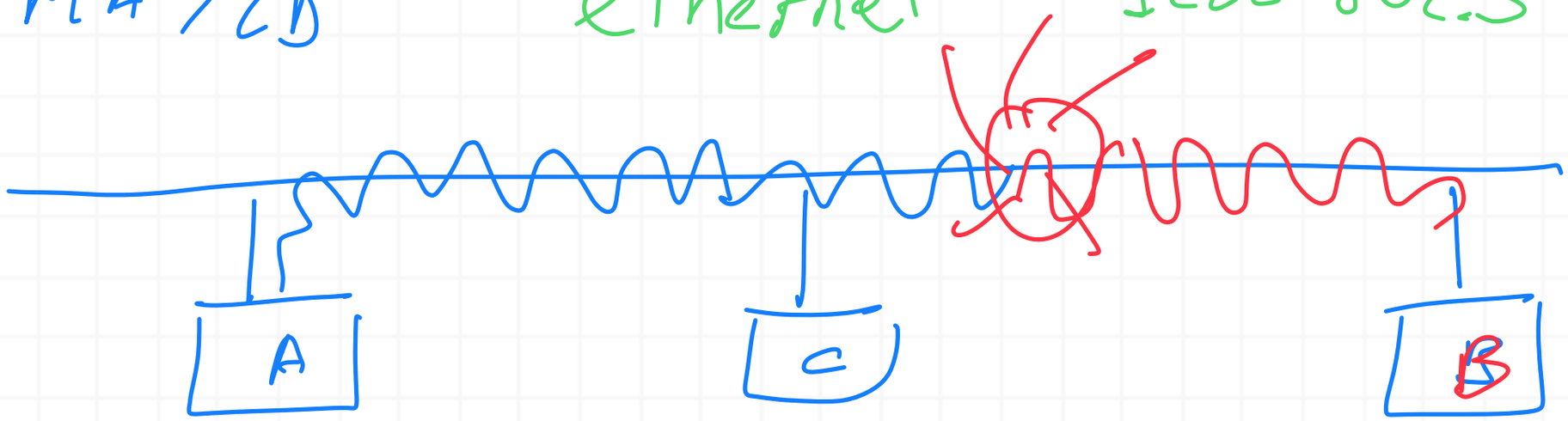



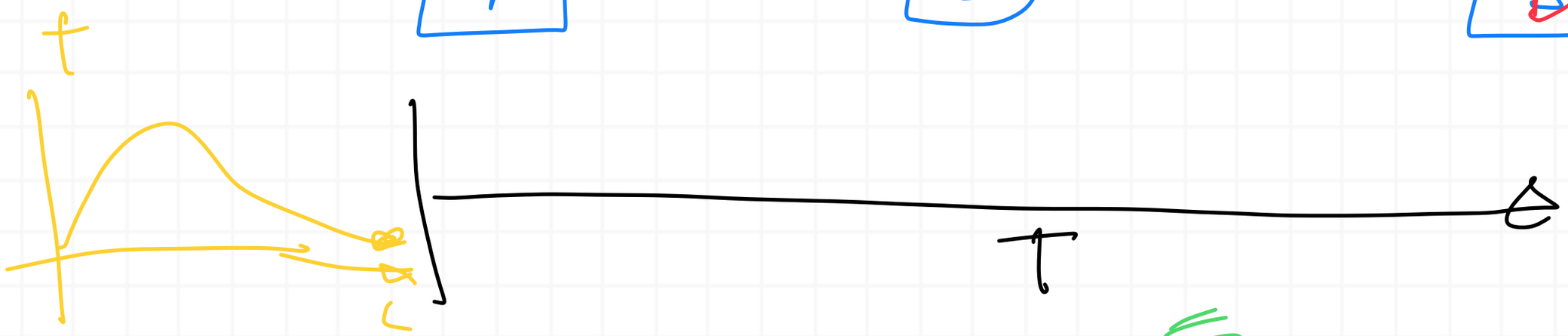



CSMA/CD

ethernet

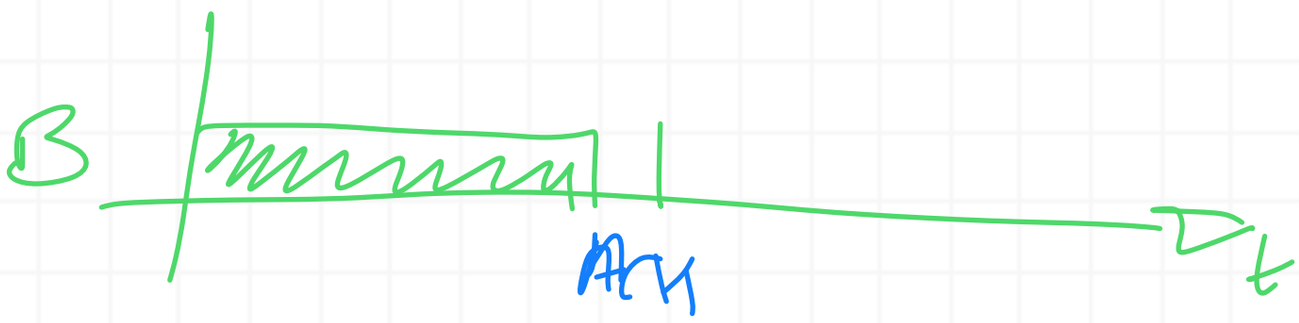
IEEE 802.3



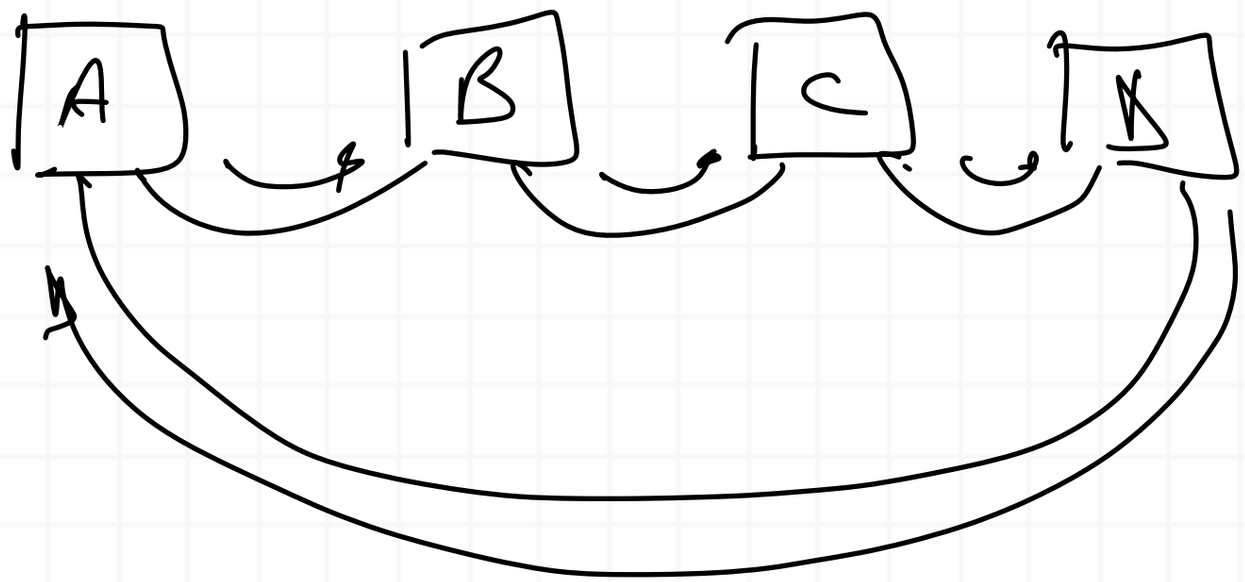




66%

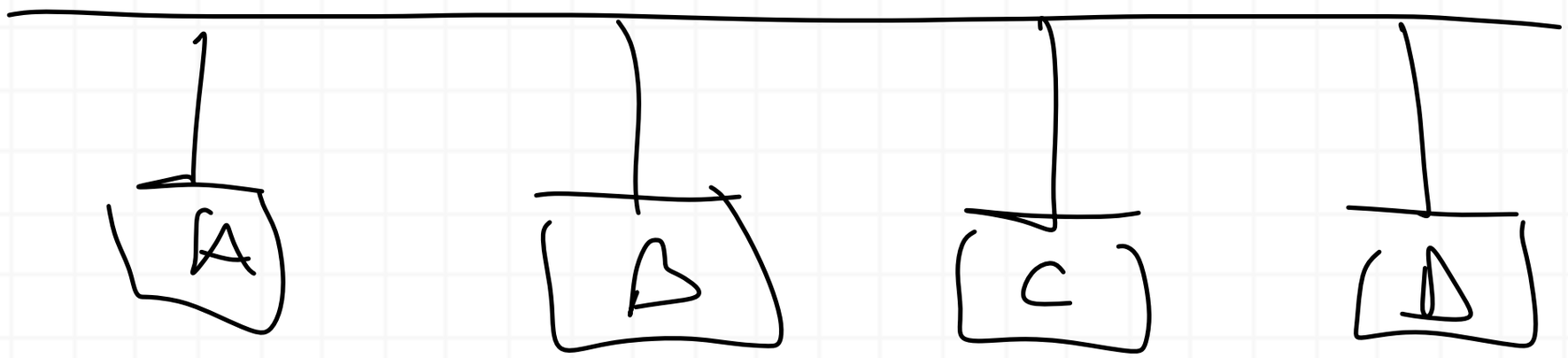
CSMA/CA



# token Ring, token Bus




$\approx 99\%$



Lectura para la próxima clase:
**Concluir el capítulo 6**